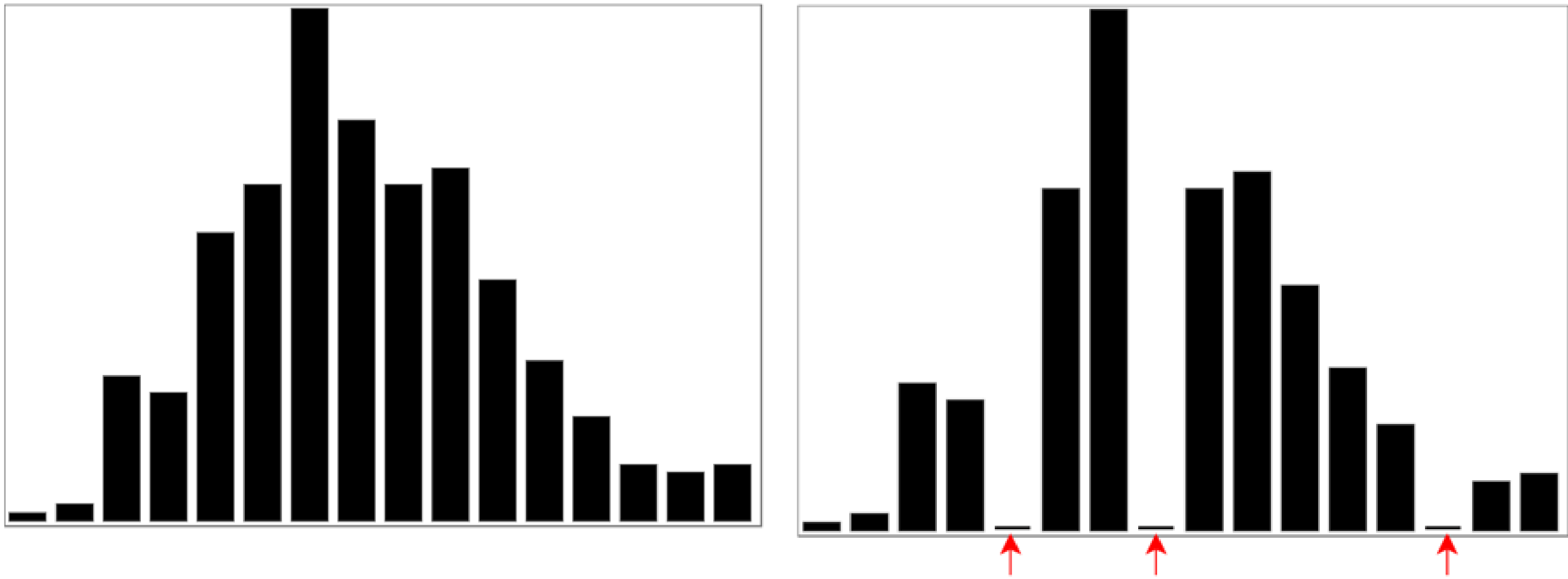




ABSTRACT

We test whether spatial visualization ability impacts recall of bar charts. We propose a methodology using a crowd-sourced online study to assess spatial visualization and measure differences in error and response times across low and high spatial individuals; we found that once a task is understood, there is no difference in response times between them. However, low spatial individuals had higher error with increased complexity and task difficulty.

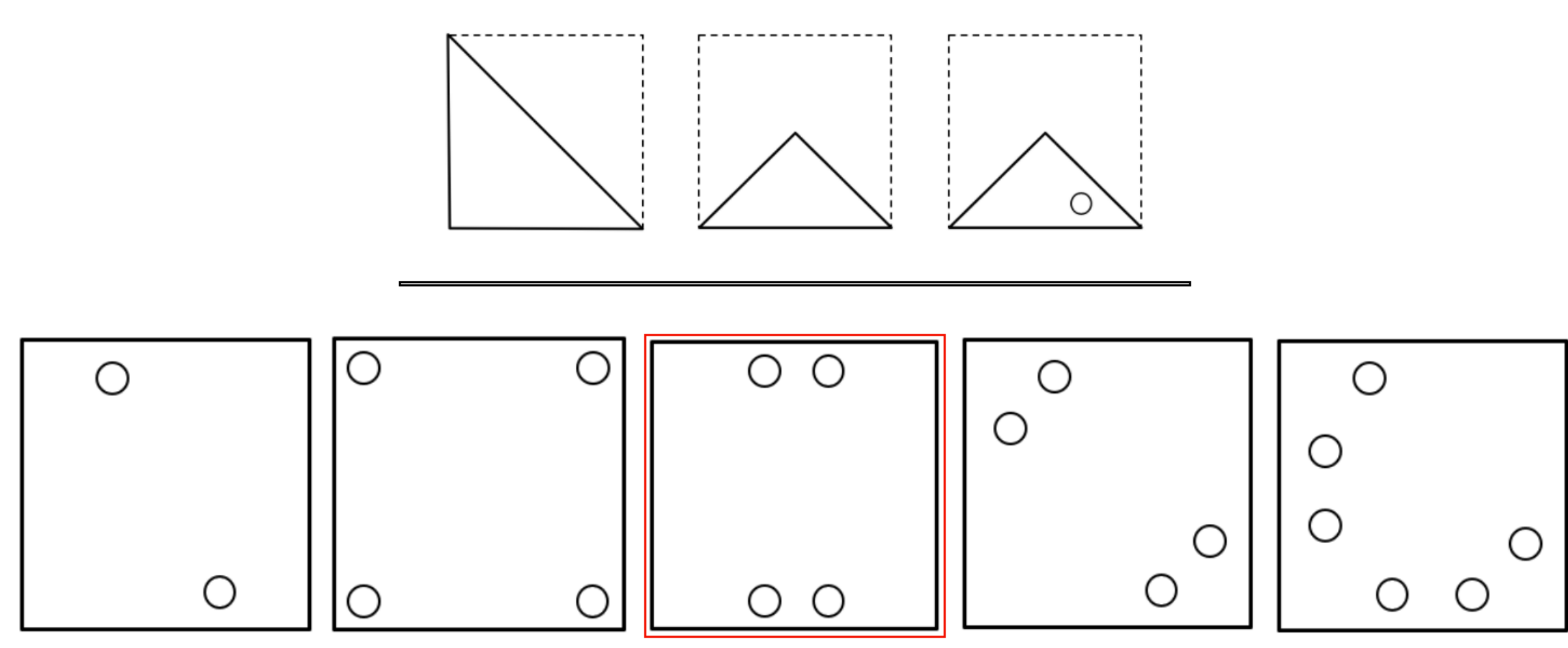
EFFECT OF SPATIAL VISUALIZATION ON RECALL OF BAR CHARTS

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SPATIAL VISUALIZATION



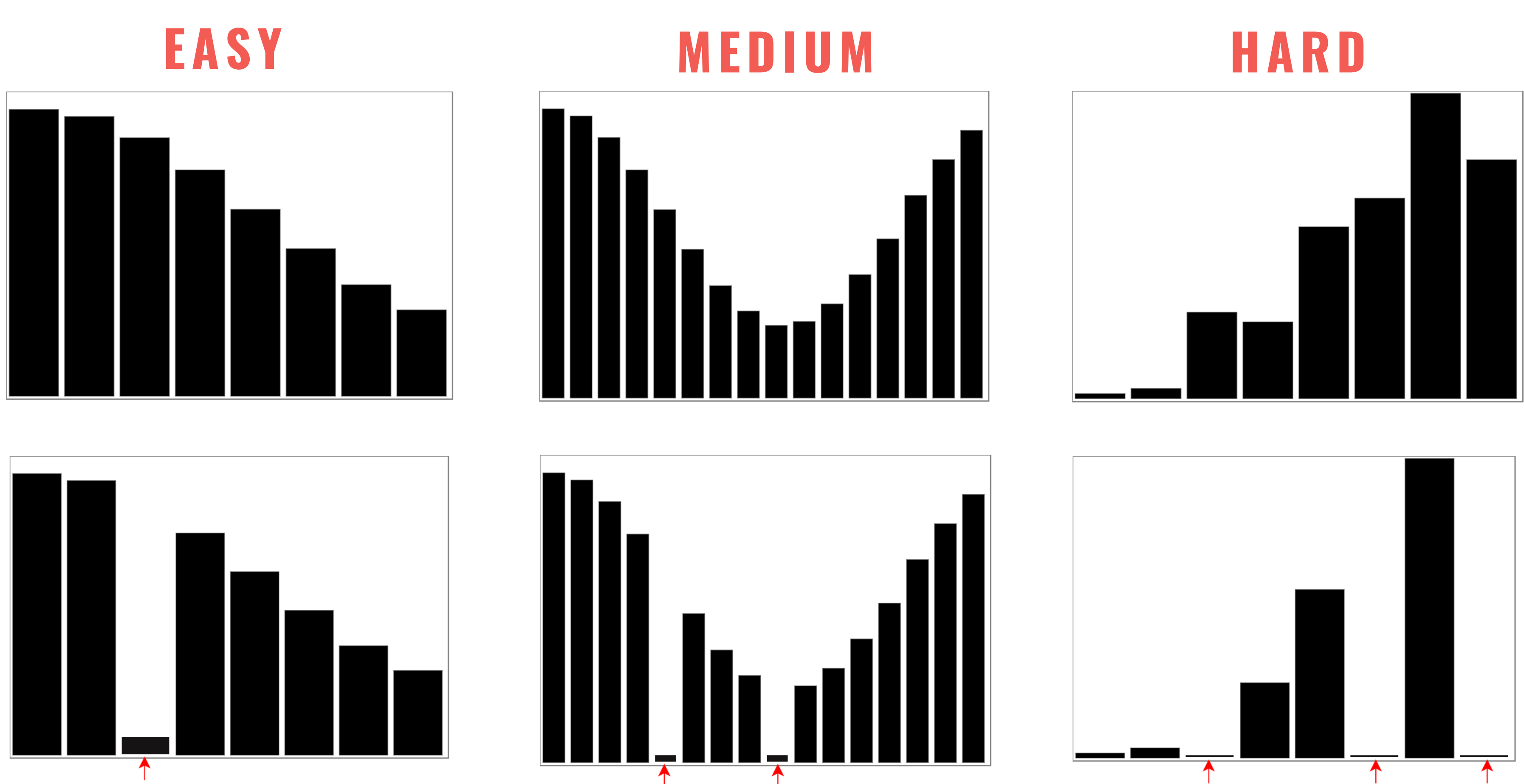
Spatial visualization involves cognitive abilities of retention, manipulation, and rotation of visual images

HYPOTHESES

RQ1: Is spatial recall of bar charts affected by spatial visualization?
RQ2: Does chart density have an effect on spatial recall?
RQ3: Does the level of task difficulty have an effect on spatial recall?
H1: Low spatial individuals will have higher response times
H2: High spatial individuals will have lower error

METHODS

Part 1: 9 Interactive bar charts: 4, 8, or 16 bars with 1, 2, or 3 spatial locations to recall. Initial viewing was limited to 5 seconds, followed by a 20 second mask and unlimited response time. Data was simulated Cosine and Poisson distributions, not associated with a specific data set. Task difficulty was defined over easy, medium, and hard depending on number of bars to recall and associated distribution.
Part 2: Psychometric spatial visualization assessment.

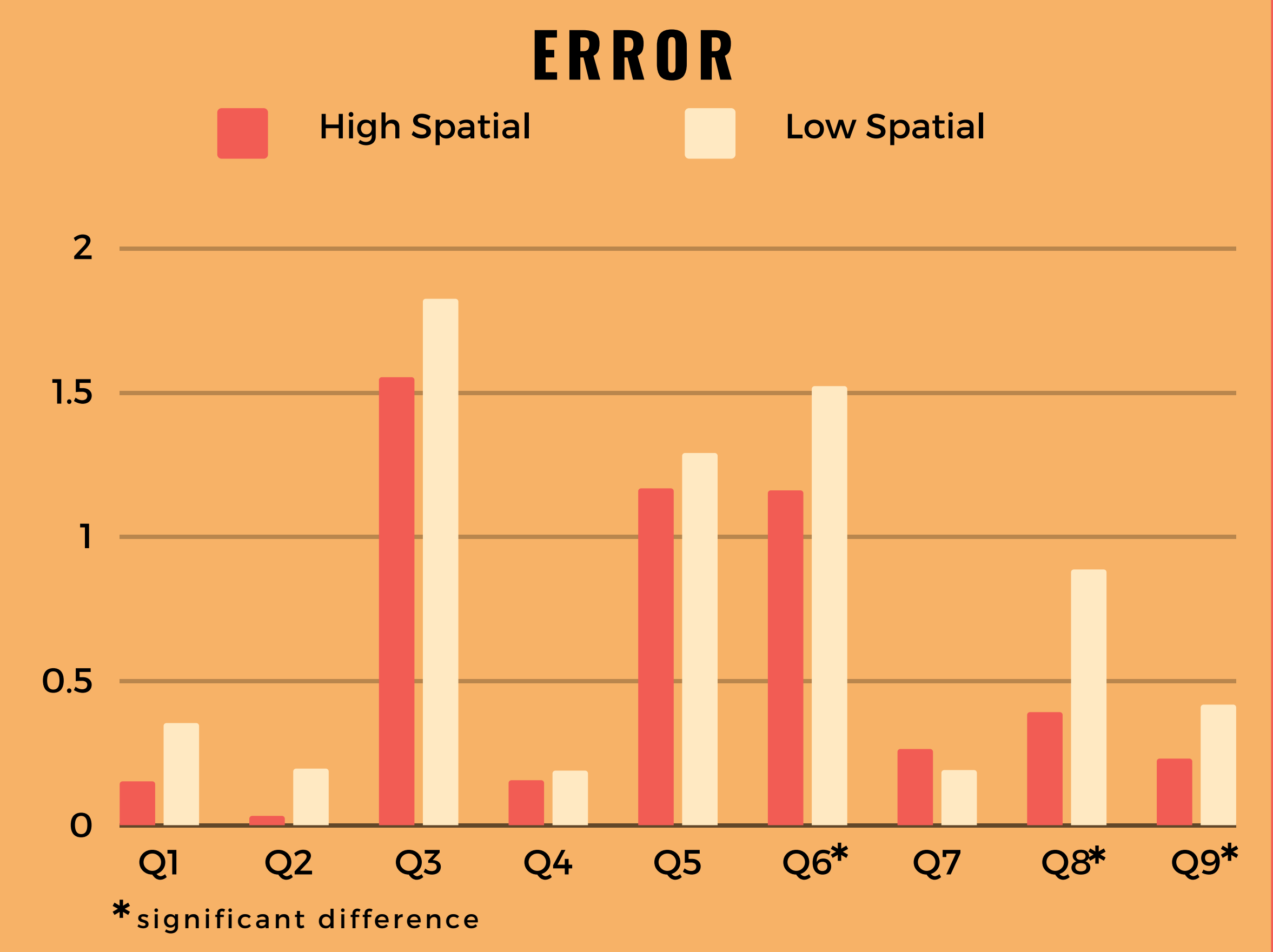


RESULTS

Participants: Out of 145 participants, 76 were classified high spatial and 69 low spatial.

Response Time: From post-hoc analysis of response time between low and high spatial individuals, there was a significant difference for stimulus 1 alone at $t=-1.87$, $p=0.06$. This suggests that once the task was understood, there is no difference in response time for low and high spatial individuals, negating **H1**.

Error: Low spatial individuals had significantly higher error on stimuli 6, 8, and 9 ($t=-2.23$, $p<0.05$; $t=-2.05$, $p<0.05$; $t=-1.9$, $p<0.05$). Stimulus 6 is the first stimulus with 3 data points; stimuli 8 and 9 are rated as the highest difficulty. Additionally, for the Cosine distribution, low spatial individuals had significantly higher error in the first bar to recall ($t=-3.21$, $p<0.01$). These findings partially support **H2**, that high spatial individuals have lower error given task complexity and chart density



Other: Significant differences were detected in various stimuli between those who found Part 1 difficult vs those who did not, between those with a bachelor's degree vs those without, and between those living in an Urban area vs those who do not.